PAYTM BUILD FOR INDIA AI HACKATHON · DELHI TRACK 1 · MERCHANT GROWTH AI

Munim

The CFO for every merchant's AI growth agent

Paytm is giving merchants AI agents that can spend money on growth. Munim decides how much they may spend, makes every offer prove itself, and stops what does not pay.

SOUNDBOX

32 grahakon ko ₹250+ ke bill par ₹15 cashback bhejein? Kharcha ₹480 tak.

Send ₹15 cashback on ₹250+ bills to 32 customers? It costs up to ₹480.

HAAN NAHI

The agent asked. Munim checked. The owner decides.

Satyam Mishra

Live demo saksham869.github.io/munim Code github.com/saksham869/munim

PROBLEM STATEMENT

Agents now spend merchants' money. Nobody is the CFO

1.57 Cr

merchants paying for Paytm payment devices (Q1 FY27)

Paytm's Pi Sales & Marketing Agent already spends on ads for small businesses: Pi suggests the budget, the owner taps approve. Cashback to a shop's own customers is the next lever, and the riskier one.

Give an agent a budget and three things go wrong:

OVERSPEND

Cashback bigger than the margin

₹40 back on a ₹100 bill: the shop earns ₹12 and pays out ₹40.

DEADWEIGHT

Paying people who'd come anyway

51.3% of occasional shoppers return within 14 days on their own. Most cashback sent to them is wasted.

NO PROOF

One shop can't tell what worked

With comparison groups this small (6 to 9 people in the demo), a shop judging by them alone stops 45% of offers that were making money (1,000 simulated campaigns).

Card numbers: Munim's run on a synthetic kirana (live demo). Margins are already thin: a distributors' federation estimates about 2 lakh kiranas shut in a year amid quick commerce (AICPDF, 2024). Sources: Paytm Q1 FY27 results, Jul 2026; Paytm blog on Pi, Sep 2026; AI Soundbox: FoneArena, Oct 2025.

PROPOSED SOLUTION

Munim: a spending limit and a scorecard for every growth agent

1

Owner sets rules

Once, by voice on the Paytm AI Soundbox or in the app: ₹500 a week, keep 6% a bill, ask me above ₹300.

2

Agent proposes

Any LLM today, Pi agents later, suggests an offer: who gets it, how much cashback, what minimum bill.

3

Munim decides

Allow it inside the rules. Hold big spends until the owner taps "Haan". Refuse with the exact fix.

ALLOW GATE DENY

4

Paytm runs it

A signed, single-use passport releases it. 1 in 5 customers don't get it for 14 days, so we can tell if it worked.

5

Munim reports

On day 14, in Hinglish on the Soundbox. Run again if it pays; if not, no rerun for 30 days. Unused budget returned.

SOUNDBOX

"Munim, sab band karo."

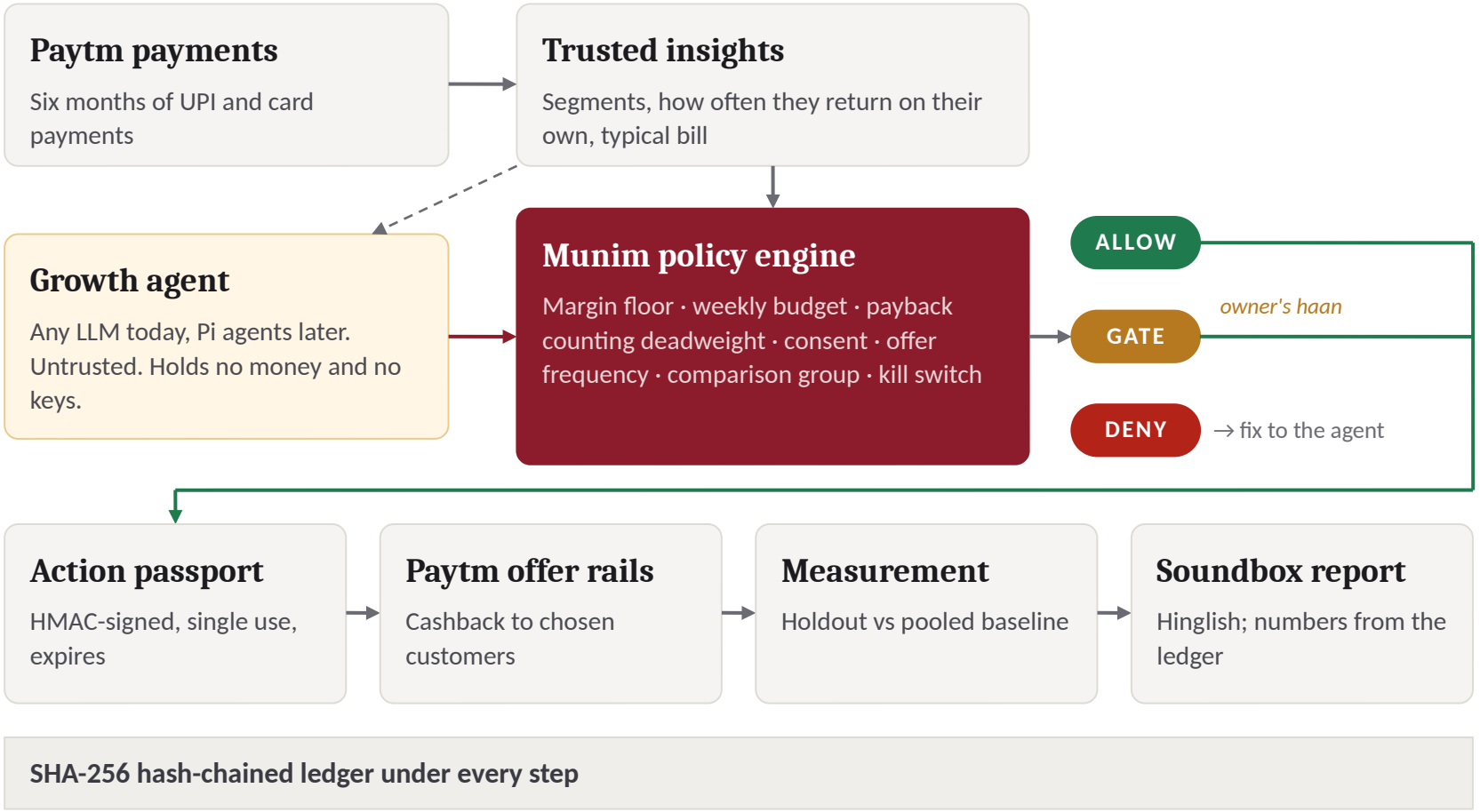
One sentence cancels running offers and holds every new one for your yes.

TAMPER-EVIDENT LEDGER

Every proposal, decision, approval, payout and result is hash-chained and sealed with Munim's key. An edit breaks the chain; a full rewrite fails the seals.

HOW IT WORKS

The agent proposes. Munim, with no LLM inside, decides



GUARANTEES

The agent never touches money or keys.

Insights come from Paytm data, never from the agent.

The agent can't pick who is held back: the split is salted with a secret.

Any error inside Munim means no.

Soundbox numbers come from the ledger, never from the model.

DEMO: ONE RUN, EVERY NUMBER FROM THE ENGINE

Six decisions on a synthetic kirana in Ghaziabad

Sharma General Store (synthetic): 650 customers, 13,772 payments in 6 months. 48 regulars stopped coming; they used to spend about ₹1,28,000 a month.

DENY #1

Agent: ₹40 back on any bill over ₹100, nobody held back

4 reasons, each with a fix: at most ₹6 at that bill (try ₹15 on ₹250+) · hold back 20% · ₹1,280 is over the ₹500 budget · loses money even if all return

GATE #2

Replan: ₹15 back on ₹250+, 32 customers, 9 held back

Worst case ₹480 is above the ₹300 limit. Owner taps "Haan"; Munim re-checks, then releases it.

DENY #3

Agent: ₹12 back for occasional shoppers

51.3% come back anyway; breaking even needs 29.0 pp more. Most it can be: ₹10.

ALLOW #4

₹10 back on ₹200+, 22 customers

Inside every rule (worst case ₹220), so it runs without asking. It is still measured.

STOPPED #5

Day 21: the ₹10 nudge

₹110 spent, 12% chance it made money. Verdict: stop. The agent can't rerun it for 30 days; ₹110 back in the budget.

GATE #6

Kill switch: "Munim, sab band karo"

The agent's next offer (worst case ₹120) would have been allowed. Now it waits for the owner.

Ledger: 20 entries, intact. Change one verdict and the chain breaks at entry 17.

Replay it yourself saksham869.github.io/munim

PROOF, NOT CLAIMS

A shop's comparison group is too small. A pooled one is not

Same run, two ways to judge it

Offer	Truth (simulated)	Shop's own group	Munim, pooled baseline
Win back regulars	Works: +20 pp	Stop it (6% chance it paid) ×	**Keep measuring (56% chance it paid) ✓**
Nudge occasional shoppers	Does nothing	Keep it (90% chance it paid) ×	**Stop it (12% chance it paid) ✓**

One run, picked to illustrate: by chance 5 of 9 held-back regulars came back anyway, and 0 of 6 occasional shoppers did. The 1,000-campaign test is the evidence.

75% right keep/stop calls (shop's own group: 68%)

24% of profitable offers wrongly stopped (shop's own group: 45%)

1,000 simulated campaigns with known truth; pooling also won on 5 of 5 more random seeds. Honest costs: it keeps slightly more losing offers (25% vs 23%), and a pooled rate 5 points too low (a festival fortnight, say) erases the gain, so the rate must come from recent, comparable history.

Expected profit per campaign (₹)



The value of the keep-or-stop call on each offer's next run, when 40% of offers do nothing.

TECHNOLOGY / TECH STACK

Small, deterministic, and ready to plug into Paytm

WHAT'S BUILT

Policy engine	Python, integer paise and basis points, no dependencies, fails closed
Measurement	Beta-binomial posteriors, pooled baselines, profit-probability verdicts
Controls	HMAC-SHA256 passports, race-safe weekly budget, kill switch
Ledger	SHA-256 hash chain, verified in one call
API	FastAPI; boots on a synthetic kirana with zero setup
LLM adapter	Any OpenAI-compatible endpoint; output parsed strictly, never trusted
Live demo	Same rules mirrored in JavaScript, parity-tested against the Python engine
Tests	180 automated tests (pytest)

PLUGS INTO PAYTM

AI Soundbox

Voice and touchscreen, 11 languages; the app for older boxes

Pi Agents

Today Pi suggests ad budgets; Munim checks the offers step

Payments data

Segments, return rates, typical bills

Offer rails

Cashback to Paytm payers who opted in (mock in demo)

WhatsApp Business

Already connected by Pi's marketing agent for leads

USP

Guardrails stop bad prompts. Munim stops bad spending

1

Economic authority

Payment controls like Google's AP2 cap amount, payee and time; Pi asks the owner to approve a budget. None checks margin, deadweight or a comparison group.

2

Deadweight-aware

Counts cashback paid to customers who'd come anyway, and refuses offers that can't pay back.

3

Proof by default

Every offer keeps a comparison group, and the agent cannot choose who is in it.

4

Pooled baselines

Turn a 30-customer campaign into a decision: the shop's own history where it has enough, similar Paytm shops where it doesn't.

5

Safe by construction

No LLM in the money path. Errors mean no. Every decision in a tamper-evident ledger.

OWNER'S APP

Agent ka ₹40 wala offer roka: ismein ghaata tha. ₹250+ par ₹15 wala offer aapki manzoori ke liye taiyaar.

The agent's ₹40 offer was refused: it would lose money. A ₹15 offer on ₹250+ bills is ready for your OK.

Refusals and margins go to the app, never out loud at the counter.

IMPACT & BENEFITS

Growth that merchants, Paytm and customers can all trust

MERCHANTS

Growth without the guesswork

- A hard weekly cap (₹500 in the demo)
- At least 6% kept on a bill at your usual margin
- Results in Hinglish when each offer ends
- One-sentence kill switch

PAYTM

Agents merchants keep paying for

- Measured lift that makes Pi Agents worth paying for
- Fewer "the agent wasted my money" escalations
- Every campaign sharpens the network baseline
- A safe path to agent-to-agent payments

CUSTOMERS

Offers, not spam

- Only customers who opted in
- At most 2 offers in 30 days
- Offers sized so the shop stays healthy
- Hinglish messages, clear terms

₹1,280 → ₹480

Worst case of the agent's first offer vs the replanned offer the owner approved; the first would lose money even if every customer came back. In simulation: ₹15.85 per campaign vs -₹17.85 keeping every offer.

BUSINESS MODEL (PROPOSED)

Part of Pi Agents, so agent-run offers are safe to sell

Bundled into Pi

Munim would ship inside Pi Agents for small businesses. Merchants pay for agent-run offers; the checks come with them.

What Paytm gets

Cashback offers it can sell to a kirana, with the owner's weekly budget as the worst case. The cashback moves on Paytm's rails.

Outcome fee, later

Optional: a small share of measured net gain. The keep-or-stop rules are fixed before an offer runs, and the owner can recheck every verdict in the ledger.

Pilot before pricing

50 kiranas for 8 weeks with Paytm, measuring reachable customers per shop, how often owners tap Haan, and the risky spend Munim refused.

SCALE, ILLUSTRATED

₹18.6 Cr

a year if 1% of the 1.57 Cr merchants on Paytm devices paid ₹99 a month for agent-run offers

1.57 lakh × ₹99 × 12 ≈ ₹18.6 Cr

An illustration, not a forecast. The ₹99 price and the 1% uptake are assumptions for the pilot to test.

Running cost: near zero. Decisions are arithmetic, with no model call.

Let the agent grow the shop.
Munim keeps the money safe.

BUILT AND RUNNING TODAY

- ✓ Policy engine, measurement, passports, sealed ledger
- ✓ 180 automated tests
- ✓ 1,000-campaign evaluation with robustness checks
- ✓ Live demo, API, import of your own payments

NEXT, FOR 31 OCTOBER

- → Live LLM agent proposing in the demo
- → The shop's own history in the measurement
- → Hindi voice for rules and "sab band karo"
- → Approval screen for the owner

With Paytm, after the event: real cohorts, offer rails, consent.

Thank you

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Live demo saksham869.github.io/munim Code github.com/saksham869/munim Email mishrasatyam3456@gmail.com

Demo data is synthetic. The offer API is a mock. Munim is an independent hackathon project, not a Paytm product.